

Quiz 1 Formula Sheet

1 Set Relations and Event Rules

1.1 De Morgan's Laws

De Morgan's laws relate the complements of sets through unions and intersections. For any two events A and B :

$$\begin{aligned}(A \cup B)^c &= A^c \cap B^c \\ (A \cap B)^c &= A^c \cup B^c\end{aligned}$$

For a collection of events $A_1, A_2, \dots, A_n$:

$$\left(\bigcup_{i=1}^n A_i \right)^c = \bigcap_{i=1}^n A_i^c \quad \text{and} \quad \left(\bigcap_{i=1}^n A_i \right)^c = \bigcup_{i=1}^n A_i^c$$

1.2 Difference Rule

The difference rule expresses the probability of one event occurring without another. For any two events A and B :

$$P(A \cap B^c) = P(A \setminus B) = P(A) - P(A \cap B)$$

1.3 Inclusion-Exclusion Principle

The inclusion-exclusion principle finds the probability of the union of multiple events by adding individual probabilities and correcting for overlapping intersections.

- **For two events:**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- **For three events:**

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

2 Conditional Probability and Rules

2.1 Multiplication Rule

The multiplication rule calculates the probability of the joint occurrence of multiple events.

- **For two events:**

$$P(A \cap B) = P(A | B)P(B) = P(B | A)P(A)$$

- **General chain rule for n events:**

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1)P(A_2 | A_1)P(A_3 | A_1 \cap A_2) \dots P(A_n | A_1 \cap \dots \cap A_{n-1})$$

2.2 Bayes' Rule

Bayes' rule updates the prior probability of an event based on new evidence or conditions. For two events A and B where $P(B) > 0$:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

Using the law of total probability, if $A_1, A_2, \dots, A_n$ form a partition of the sample space:

$$P(A_k | B) = \frac{P(B | A_k)P(A_k)}{\sum_{i=1}^n P(B | A_i)P(A_i)}$$

3 Independence and Exclusivity

3.1 Mutually Exclusive Events

A collection of events $A_1, A_2, \dots, A_n$ is ****mutually exclusive**** (or disjoint) if no two events can occur at the same time.

$$A_i \cap A_j = \emptyset \quad \text{for all } i \neq j$$

Consequently, the probability of their intersection is zero, and the probability of their union simplifies to:

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i)$$

3.2 Pairwise Independent Events

A collection of events $A_1, A_2, \dots, A_n$ is ****pairwise independent**** if every possible pair of events is independent of each other. For all $i \neq j$:

$$P(A_i \cap A_j) = P(A_i)P(A_j)$$

3.3 Mutually Independent Events

A collection of events $A_1, A_2, \dots, A_n$ is ****mutually independent**** (or collectively independent) if the multiplication rule holds for ***every*** sub-collection of events. For any subset of indices $I \subseteq \{1, 2, \dots, n\}$ where $|I| \geq 2$:

$$P\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} P(A_i)$$

Note: Mutual independence implies pairwise independence, but pairwise independence does not guarantee mutual independence.